

by stray light, for example, the system increases weighting on inertial and sun sensors for a defined interval.

Mission designers specify acceptable uncertainty budgets for each phase. Close-proximity operations demand tight bounds and aggressive fault handling, whereas cruise phases tolerate looser limits. The navigation manager enforces these policies by switching estimator configurations, increasing sample rates, or requesting ground verification when thresholds are exceeded.

Testing, simulation, and validation

We cannot validate autonomous systems using only nominal test cases. System-level simulations must inject faults, correlated failures, and environmental perturbations.

Hardware-in-the-loop platforms exercise flight hardware under realistic timing and bus loads while emulating sensor faults. Testbeds replay recorded orbital imagery and synthesise plausible fault progressions to evaluate both detection sensitivity and avoidance of false positives.

Formal verification is especially important for low-level emergency logic that commands thrusters or power switching. Where full verification is infeasible, layered redundancy and human-in-the-loop checks provide practical risk mitigation.

The research frontier and near-term future

Current research extends autonomy through cooperative constellations that share navigation and health data for distributed fusion and collective diagnosis. Agent-based FDIR allows satellites to redistribute responsibilities after failures.

Integrated photonic sensors and chip-scale inertial units promise further reductions in size, weight, and power while maintaining high performance. At the algorithmic

level, explainable and certifiable machine learning improves transparency and safety for critical applications.

Programmatic importance

Robust autonomous navigation improves mission safety, reduces operational costs, and enables new mission classes. For national space programmes and industry alike, autonomy lowers barriers to deep-space science and commercial operations while supporting resilient Earth observation, lunar logistics, and exploration architectures.

From a national security perspective, onboard detection and recovery enhance survivability and responsiveness, reducing downtime and increasing operational tempo in contested environments.

Closing synthesis: How to think about integration

Autonomous fault diagnosis and adaptive multi-source fusion are not separate modules but interconnected elements of a single system. The most resilient architectures emerge when electronics design, estimation theory, data-driven detection, algorithmic fusion, and mission policy are co-designed.

Hardware enables rapid detection and safe reconfiguration. Model-based and data-driven approaches collaborate to identify anomalies. Adaptive fusion maintains accuracy under degradation. Supervisory policies align autonomy with mission priorities.

Historic demonstrations such as AutoNav and SEXTANT established feasibility. Advances in electronics and algorithms now provide practical capability. As radiation-tolerant computing and explainable machine learning continue to improve, spacecraft will operate farther from Earth with greater independence and confidence. The shift to autonomy is no longer optional; it is a core engineering requirement for the next era of space exploration. **EFY**

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